

Ryan D. Heberlig

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EDUCATION

Northeastern University, Boston MA – *B.S. in Mechanical Engineering* **Expected Graduation 2028**

- 3.92 GPA; Dean's List
- Related Coursework: Mechanical Computation and Design, Dynamics, Fluids, Thermodynamics

SKILLS & CERTIFICATIONS

Technical Skills: Fixture Design, GD&T, DFM & DFA, Rapid Prototyping, FEA

Software & Tools: Creo, SolidWorks, Onshape, Ansys, MATLAB, Arduino

Manufacturing: Manual Mill & Lathe, Laser Cutting & Engraving, SLA & FDM 3D Printing

Certifications: Lean Six Sigma Green Belt, CSWA, Tec-Ease GD&T Fundamentals

WORK EXPERIENCE

Amphenol PCD – *Manufacturing Engineering Co-Op* **Jan 2026-Jun 2026**

- Led integration of laser engraving to replace ink marking, establishing a standardized process, creating equipment and leading training sessions for future operators and engineers, reducing part-marking time by up to **6x** across 7 product lines.
- Designed 18 jigs, fixtures, and manufacturing tools using Creo, integrating with pneumatic and electronic systems as well as handheld tool applications.
- Developed a raceway-style power distribution system for low-pressure molding, integrating off-the-shelf components with custom-designed parts to maximize durability, reducing maintenance downtime by **70%** and increasing maximum molding capacity by **25%**.
- Prototyped a camera calibration attachment, vibratory bowl feeder system and other fixturing to enable automated part sorting for a 5-axis robotic arm for pick-and-place application.
- Repaired industrial machinery used in factory operations, diagnosing equipment failure using root cause analysis.

PROJECTS

Generate Product Development Studio – *Hardware Engineer* **Sep 2026-Present**

- Sketching and evaluating single-motor dispensing mechanisms for the travel unit of an automated pill dispenser, minimizing footprint while maintaining reliability.

Forge Product Lab – *Hardware Team Member* **Feb 2026-Apr 2026**

- Created a braille E-reader alongside a team of 6 able to translate PDF text into braille.
- Designed a six-pin actuation mechanism in Onshape for standard refreshable braille characters.
- Organized ME system meetings and led design reviews alongside EE Lead.
- Fabricated FDM and SLA 3D-printed prototypes integrated with N20 motors & other electronics.

Northeastern Electric Racing – *Member* **Sep 2024-Dec 2025**

- Designed the front anti-roll bar for the team's race car, including geometry selection, material choice, stress analysis and stiffness calculations to optimize handling and cornering performance.
- Created jigs in SolidWorks for welding suspension mounts to the chassis.
- Optimized parts using FEA, reducing weight by up to **60%** while maintaining the minimum FOS.

Punching StrengthTester – *Cornerstone of Engineering I* **Oct 2024-Dec 2024**

- Co-developed an electromechanical punching strength tester rated for **450+ N** loads.
- Designed and laser cut target, braces and electronics housing using SolidWorks.
- Integrated an accelerometer, LED lightstrip, LCD screen and UI using C++ and Arduino.

ADDITIONAL INTERESTS

Professional Interests: Product Design, Manufacturing, Rapid Prototyping

Hobbies: Volleyball, Reading, Drawing, Hiking, Travel